EL JOKER — Script du pitch, questions du jury et vidéo de démo

Version 7 · 20 octobre 2026 · Hedera Cross Campus Challenge · Team Chain619 · Rihem Ghoudi & Mariem Souayh

> Le deck suit exactement les 7 parties du guide : Project & Team, Problem, Solution, Product / PoC, Technology & Hedera, Impact & Adoption, Vision et Conclusion (sur deux slides). 11 slides, environ **5 minutes, démo comprise**. Le texte à dire est en anglais (le guide accepte aussi le français), les consignes sont en français. Tous les montants sont en dinars tunisiens.

Qui parle quand

Rihem : slides 1, 2, 5, 9 et 10. Mariem : slides 3, 4, 6, 7 et 8. La slide 11 se dit à deux.

#	Partie du guide	Slide	Qui parle	Durée	Fin à
1	Project & Team	Cover	Rihem	25 s	0:25
2	Problem	Three ways agents fail	Rihem	30 s	0:55
3	Solution	AI proposes. Rules decide. Hedera proves.	Mariem	30 s	1:25
4	Product / PoC	Request → Proof (le flow)	Mariem	20 s	1:45
5	Product / PoC	Live demo : payé en dinars	Rihem (en direct)	35 s	2:20
6	Product / PoC	Le coffre : signature humaine	Mariem (en direct)	45 s	3:05
7	Technology & Hedera	Architecture + 4 services Hedera	Mariem	35 s	3:40
8	Technology & Hedera	Why not a database	Mariem	20 s	4:00
9	Impact & Adoption	Works today. Who needs it.	Rihem	25 s	4:25
10	Vision	Every AI agent that spends money, under control.	Rihem	20 s	4:45
11	Conclusion	Autonomous, but never uncontrolled.	Mariem puis Rihem	20 s	5:05

Préparer la démo avant de monter sur scène

- L'application tourne (`npm start`), en mode réel : badges « Vault 2-of-2 » et « Pays in EJTND (HTS) » en haut.
- Lancez le scénario **A** avant le pitch : sur scène, Rihem montre la mission déjà payée, sans attente.
- Lancez le scénario **I** avant le pitch : il s'arrête sur « Human approval required ». Sur scène, Mariem tape la raison et le PIN, puis clique Approve : Hedera paie en quelques secondes. C'est le moment fort.
- Ouvrez HashScan dans un autre onglet.
- Si la démo plante : « Here is the same run, recorded earlier on testnet. »

Ce que vous dites

Pour chaque slide : « Say » = ce que vous dites (5 minutes au total). « Details » = pour comprendre et répondre au jury, à ne pas lire sur scène. Les mêmes notes sont dans le deck.

1. Project & Team — 20 s — Rihem.

> « Hi, we are Chain619: I'm Rihem, and this is Mariem. Our world is moving forward, and AI is taking a huge place in every technology. But we still doubt it, and we can't give it access to our wallets. So we built EL JOKER, "5aliha 3lina", leave it to us: a team of five agents that buy things for an organization, under control. Every step is proven on Hedera. »

> **Details :** EL JOKER is the control layer for AI agents that spend money. One AI (Claude) proposes; four strict programs verify, decide and pay. Above the organization's limit, the money waits in a Hedera vault that only a human co-signature can open. Stack: a multi-agent system, Claude (Anthropic), the Hedera Consensus Service, scheduled transactions (the 2-of-2 vault), the Token Service (our test dinar EJTND) and a Mirror Node verifier.

2. Problem — 30 s — Rihem.

> « Why don't we trust AI with money? Because AI agents can fail in many ways. They hallucinate and invent things, they overspend or pay twice, and their logs can be changed. So the question is: who controls them, and how do we prove what they did? Today you must choose: autonomy or control. We give you both. Mariem? »

> **Details :** Hallucinate: language models produce fluent but unverified facts; an agent can invent a supplier, a price or an approval and act on it with real money (e.g. a "verified" broker with no license). Overspend: agents optimise the goal, not the budget; retries, timeouts and race conditions turn one purchase into duplicate payments (e.g. the same invoice paid twice after a network retry). Hide the trail: logs live in a database the operator controls; they can be edited or deleted after the fact, so there is no independent audit trail. Why it matters: an NGO must justify every dinar to its donors; a company must respect its spending policy and pass its audits.

3. Solution — 30 s — Mariem.

> « Our rule: AI proposes, rules decide, Hedera proves. Meet the crew. Lem3alem, the coordinator, turns the request into a clear mission. Lemhaf is the only AI: Claude compares the offers. Nasnes verifies, 3ases decides with fixed rules, and Kanawita pays. Claude never touches the money. Above the limit, a human must sign, on Hedera. »

> **Details :** It is a separation of duties, like in a finance team: the one who chooses is never the one who approves or pays. Lem3alem: free text becomes a typed, schema-validated mission (who, where, what, budget); missing facts are asked, never guessed. Lemhaf: Claude ranks the marketplace offers and explains the trade-offs, with a strict JSON output; it can only recommend. Nasnes: 10 independent checks (supplier registry, license, evidence hashes, invoice reuse, price sanity, the AI's claims). 3ases: 13 deterministic rules from the organization's policy plus a risk score: approve, block or escalate. It has no LLM, so nobody can talk it into a yes. Kanawita: pays only after the approval is anchored on Hedera, with an idempotency key. Control: above the limit, a human co-signs on Hedera.

4. Product / PoC : le flow — 20 s — Mariem.

> « Here is the flow of one mission: request, plan, offers, checks, rules, payment. Each step leaves a proof on Hedera. Above the limit, the money waits for a human signature. Rihem, show them. »

Details : At every step the agent writes its decision to the database (SQLite). EL JOKER hashes it (SHA-256 of canonical JSON), chains it to the previous hash and publishes the fingerprint on the Hedera Consensus Service (topic 0.0.11007004). Kanawita waits until the approval proof reaches consensus before paying. Above the limit, 3ases escalates instead of approving: Kanawita creates a scheduled transfer from the vault, signed by the agents' key (1 of 2), and only the human approver's signature (2 of 2) releases it.

5. Product / PoC : la démo — 35 s — Rihem, live: mission A already run before the pitch.

« Here is a real run on Hedera testnet. Because of a water crisis in Gabès, an NGO requested water and medicine for 500 people within 48 hours, with a budget of 15,000 dinars. The flow goes like this: Claude ranks 3 offers and picks Oasis, 12,450 dinars. Nasnes passes 10 checks out of 10, 3ases runs its rules and says yes, and finally Kanawita pays 12,450 EJTND, our test dinar. One proof per decision: 6 decisions anchored on Hedera and confirmed by the Mirror Node. A fake supplier or a double payment: blocked. Mariem? »

Details : Oasis delivers in 20 h; 3ases passes 12 of 12 rules, risk LOW, NGO policy 2.1.0 (autonomous limit 15,000 TND). The payment is an HTS TransferTransaction of 12,450 EJTND (1 token = 1 dinar, a test token with no real value), receipt SUCCESS, confirmed on the Mirror Node, with a HashScan link. Exactly one payment: an idempotency key plus a unique database rule; a replay gets HTTP 409. Refusals (budget too small, unverified supplier, recycled invoice) are proven on Hedera too.

6. Product / PoC : le coffre — 45 s — Mariem, live: mission I already waiting for approval.

« Now a company, ACME: 35,460 dinars of laptops. Its limit is 6,000. So Kanawita can't pay. It locks the money on Hedera: one signature out of two. I approve with my PIN: the second signature. Now Hedera pays by itself. If I refuse, nothing moves. »

Details : ACME policy 1.1.0: autonomous limit 6,000 TND. The vault account's key is a KeyList with threshold 2 (the agents' key + the approver's key). Kanawita sends a ScheduleCreateTransaction (vault → supplier, 35,460 EJTND) signed by the agents: 1 of 2, visible as pending on HashScan. Operator + reason + PIN: the approver adds a ScheduleSignTransaction, 2 of 2, and Hedera executes the transfer itself. A rejection sends a ScheduleDeleteTransaction: the money never leaves the vault. The approval is bound to the input hash of the exact facts reviewed: change one fact and the approval is refused. Even if our server were hacked, its key is only one of two. In the demo the server keeps the approver key behind the PIN; in production it lives in the approver's own wallet.

7. Technology & Hedera — 35 s — Mariem.

« Our architecture: a web app, an API with five agents, Claude for only one of them, and Hedera. We use four Hedera services. Consensus Service: a fingerprint of every decision. Scheduled transactions: the two-signature vault, so even our server can't move big amounts alone. Token Service: our test dinar. Mirror Node: anyone can check. Only hashes go on-chain. »

Details : No smart contract to audit: only native services. HCS: SHA-256 of canonical JSON, chained, sent with TopicMessageSubmitTransaction, ordered and timestamped (topic 0.0.11007004). Schedules: KeyList threshold 2, ScheduleCreate then ScheduleSign, then Hedera executes; ScheduleDelete on a rejection (vault 0.0.11090535). HTS: token EJTND 0.0.11090536, 3 decimals like millimes, fixed supply, no admin key, so nobody can mint more. Mirror Node: a free REST history; the public verifier (/verify) reads it from the browser and recomputes every hash with Web Crypto. Architecture: Next.js web app → Express API with 5 agents and SQLite → Hedera SDK → Mirror Node → public verifier. Fees are fixed in USD: about one cent per mission; finality in a few seconds.

8. Technology & Hedera : why not a database — 20 s — Mariem.

> « Why not a database? A database can be changed. Someone changes the price and every hash: locally it looks fine, Hedera says tampering detected. Anyone can check it in their own browser. Rihem? »

> **Details :** Tested live: editing the approved price is caught even locally, but an insider who also recomputes every hash, or deletes the payment record, fools a local check. Hedera catches all three. How it works: (1) each event becomes canonical JSON, hashed with SHA-256 together with the previous hash: a chain; (2) the fingerprint is published on HCS: ordered, timestamped, final; (3) verify = recompute every hash from the database and compare it with the copy on the Mirror Node: one byte changed means TAMPERING DETECTED. The public verifier does this in any browser, without any access to our servers.

9. Impact & Adoption — 25 s — Rihem.

> « This works today: nine scenarios passed live on Hedera testnet, 115 automated tests. Who needs it: NGOs that must justify every dinar to donors, then companies that put agents in procurement. Donors and auditors can check any mission themselves. Our model: a subscription per organization. To be honest: suppliers and the dinar token are demo. »

> **Details :** The live check passed 11/11 (the 9 scenarios A–I, deleted-event detection, the independent public verifier) with real Claude, the vault and EJTND. 115 automated tests, including 7 in a real browser. About one cent of Hedera fees per mission. Who and why: NGOs and grants (a public, tamper-proof audit trail per mission); companies (limits enforced by the ledger, faster purchases); donors and auditors (audits in minutes, without access to our servers); agent platforms (rules, a human signature and proof as a service). Business model (hypothesis, not validated): a SaaS subscription per organization plus a small fee per anchored proof. Honest limits: fictional suppliers, a test dinar token, an approver key held by our server, a demo PIN.

10. Vision — 20 s — Rihem.

> « Our vision: millions of AI agents will spend money, and each one should pass through rules, a human for big amounts, and public proof. Next: real suppliers, approvers with their own wallets, two pilots in Tunisia, then policies for every sector. Starting in Tunisia, on Hedera. »

> **Details :** Now, pilot-ready: real suppliers and registries, and a licensed partner for a real dinar token (it needs the central bank's approval). Next, keys in wallets: approvers sign from their own wallet (for example HashPack), so our server holds only one key of two. Then, two pilots in Tunisia: one NGO and one company, real purchases, measured results (time saved, fraud blocked). Scale: a policy library per sector, a shared supplier registry and an SDK to put EL JOKER in front of any agent platform. Our goal: the trust layer for agentic payments on Hedera.

11. Conclusion — 20 s.

> « Mariem: To conclude. AI agents now spend real money. With EL JOKER, the AI never decides, Hedera enforces the human signature, and anyone can check. Next: two pilots in Tunisia. — Rihem: EL JOKER: autonomous, but never uncontrolled. Thank you. »

> **Details :** Why it matters: agents now spend real money; without hard limits and proof, no NGO or company can trust them. What is different: the AI never decides; fixed rules decide, Hedera enforces the human signature above the limit, and anyone can verify every step. What is next: two pilots in Tunisia, approvers with their own wallets, a real dinar partner, then any agent and any sector. Links on the slide: GitHub, the HashScan topic, and /verify for any mission.

Questions du jury

Une ou deux phrases maximum. Si vous ne savez pas : « Good question, we haven't validated that yet. » Mariem répond à l'IA et à Hedera ; Rihem au reste.

Le projet

Isn't this only for NGOs? (Rihem)
« No. You saw a company too. Any organization that lets an agent spend needs this. »

Only one agent uses AI. Why? (Mariem)
« The hard part is making AI safe with money. AI suggests; code decides. »

Hedera

Why Hedera and not a database? (Mariem)
« A database can be changed, even by us. Hedera can't, and it enforces the human signature. »

How does the vault work? (Mariem)
« It needs two keys: ours and a human's. We prepare the payment; only the human's signature releases it. »

Who holds the approver key? (Mariem)
« In the demo, our server, used only after the PIN. In real life, the approver's own wallet. »

Is your dinar token real? (Rihem)
« It's a test token: one token equals one dinar. A real one needs a licensed partner and the central bank's approval. »

Why not put everything on-chain? (Mariem)
« Privacy. Only fingerprints go on-chain: no names, no prices. »

What if Hedera is down? (Mariem)
« Then nobody gets paid. No proof, no payment. »

Can I check without trusting you? (Mariem)
« Yes. Our public verifier reads Hedera from your browser and recomputes everything. »

Is it real or simulated? (Rihem)
« Real, on testnet. Every payment has a link on HashScan. »

L'IA et la sécurité

What if the AI is wrong or tricked? (Mariem)
« It can only suggest. Nasnes checks it, and 3ases decides. »

Can you pay twice? (Rihem)
« No: a unique payment key, database rules, and an invoice check. »

Can someone fake an approval? (Rihem)
« It needs an operator and a PIN, tied to the exact facts. And Hedera needs the human's signature. »

Marché et suite

Who pays? (Rihem)
« NGOs first, then companies: a subscription plus a small fee per proof. Not validated yet. »

Have you talked to a real NGO or company? (Rihem)
« Not yet. Our next step is two pilots in Tunisia. » *[Si vous avez parlé à quelqu'un d'ici le pitch, dites plutôt ce qui s'est vraiment passé.]*

Who are your competitors? (Rihem)
« Others give autonomy or logs. We give fixed rules, a human signature on Hedera, and public proof. »

What would you do with three more months? (Rihem)
« Real suppliers, a dinar partner, approvers' wallets, and two pilots. »

What was the hardest part? (Rihem ou Mariem)
« A lost receipt created a false tampering alarm. We fixed it and added a test. »

Vidéo de démo (3 minutes)

Six moments, phrases courtes. Rihem fait la voix des démos, Mariem celle des pièges et de Hedera. Enregistrez après avoir configuré le coffre et EJTND, pour que tout soit réel.

Temps	À l'écran	Action	Voix off
0:00–0:12	Page d'accueil	Aucune	**Rihem :** « AI agents can now spend money. EL JOKER keeps them under control. AI proposes, rules decide, Hedera proves. »
0:12–0:50	Scénario A	« Go for EL JOKER », puis Envoyer ; couper l'attente	**Rihem :** « An NGO in Gabès, 15,000 dinars. Claude picks Oasis: 12,450 dinars. Checked, approved, paid in EJTND, our test dinar on Hedera. Six proofs on Hedera. »
0:50–1:05	Panneau Actions	« Replay payment (attack) »	**Mariem :** « Same payment sent again. Refused. Still one payment. »
1:05–1:25	Puce D	Envoyer, montrer BLOCKED	**Mariem :** « The AI picks a fake supplier. Nasnes catches it. Blocked. No money moves. »
1:25–2:10	Puce I (ACME), carte d'approbation	Envoyer ; montrer « 1 of 2 signatures » sur HashScan ; raison + PIN, Approve	**Rihem :** « A company: 35,460 dinars of laptops. Its limit is 6,000. The money is locked on Hedera. I sign with my PIN. Now Hedera pays. »
2:10–2:50	« Proof & verify », puis le vérificateur public	« Insider rewrites hashes too », puis « Open the public verifier »	**Mariem :** « Someone changes the price and every hash. Hedera says no: tampering detected. Anyone can check. »
2:50–3:00	Slide finale	Aucune	**Les deux :** « EL JOKER: autonomous, but never uncontrolled. »

Le bouton de falsification reste grisé tant que les preuves ne sont pas ancrées : attendez quelques secondes.

Avant d'enregistrer

☐ Lancer `npm run hedera:setup -- --vault --stablecoin` dans ton terminal, puis redémarrer l'application.
☐ Vérifier les badges : « AI · claude-opus-5-5 », « Hedera testnet », « Vault 2-of-2 », « Pays in EJTND (HTS) ».
☐ Lancer `npm run demo:check` : tout doit être PASS.
☐ Vérifier qu'il reste des HBAR testnet et des crédits Anthropic.
☐ Faire une répétition sans enregistrer.
☐ Plein écran 1920×1080, Ne pas déranger activé.
☐ Ne jamais montrer le fichier .env ni un terminal avec des clés.
☐ Enregistrer avec Xbox Game Bar (Win + Alt + R) ou OBS, dans une pièce calme.
☐ Au montage : couper les attentes, zoomer sur « 1 of 2 signatures » et TAMPERING DETECTED, sous-titres en anglais.
☐ Exporter en MP4 1080p ; vérifier la durée maximale du hackathon.